

WASP-2b

R-1765

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_130612 · created 2026-08-02T21:27:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456455.8148 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.031
Transit depth (Rp/Rs)^2	2.36 +/- 0.96 [%]
Orbital Inclination (inc)	84.3 +/- 1.67
Airmass coefficient 1 (a1)	1.0 +/- 0.015
Airmass coefficient 2 (a2)	0.0021 +/- 0.0039
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	#2 - [188, 284]
Optimal Method	PSF photometry
Transit Duration (day)	0.064 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.153 ± 0.031 vs 0.1326 (deviation 0.66σ)
Duration fit vs published	0.064 d vs 0.0737 d (deviation 0.39σ)
Mid-time O–C	7.5 min
Depth SNR	4.9
Residual scatter	1.49 %

Light curve

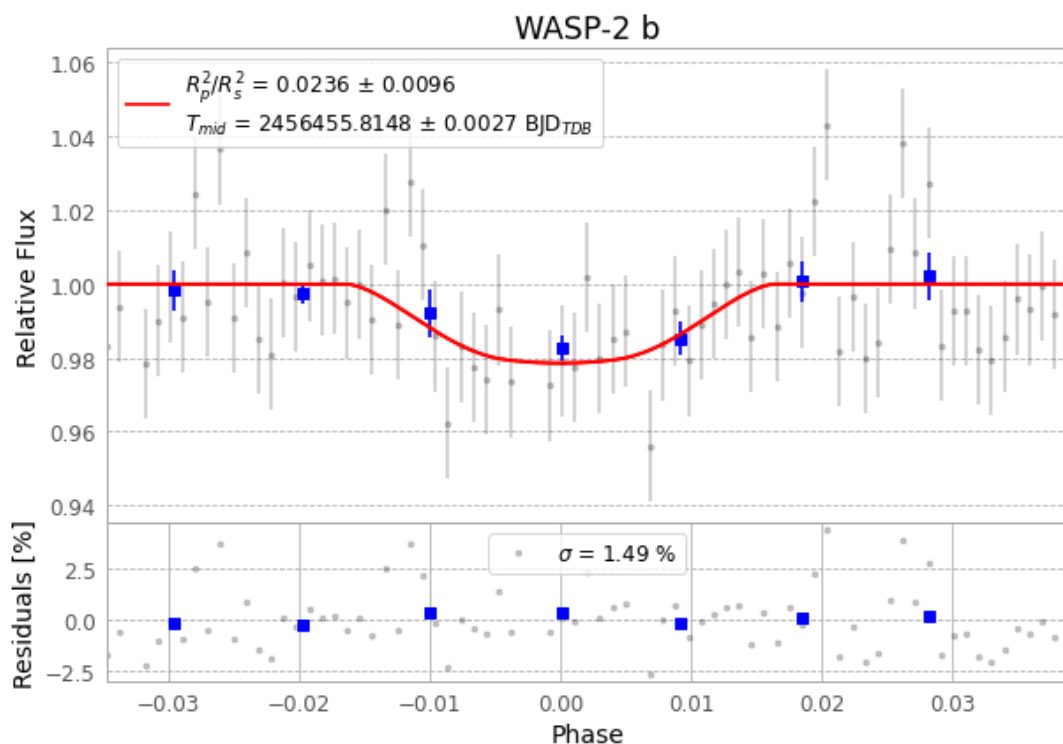


Figure 1 — FinalLightCurve_WASP-2 b_2013-06-11.png (SHA-256 d82efe683fb8e20f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [145, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 407fb4ac706f2cbd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2130612053913.FITS → WASP-2130612092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130612.log (4ec052cb786f...), FinalLightCurve_WASP-2 b_2013-06-11.png (d82efe683fb8...), FinalLightCurve_WASP-2 b_2013-06-11.csv (8a7f3ac3c37f...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:27Z.